

Derivatives from Tables

Directions: Use each table to find the indicated derivative. Show all work and simplify your answer.

1. $h(x) = f(x)g(x), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-4	2	2	1
2	-1	2	1	-5
3	-4	6	-1	4
6	8	1	-4	-4

$$h'(2) = f(2)g'(2) + f'(2)g(2) = (-1)(-5) + (1)(2) = \boxed{7}$$

2. $h(x) = \frac{f(x)}{g(x)}, h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	6	-4	2	3
2	8	2	-2	1
3	8	-1	3	-2
6	6	4	-2	-5

$$h'(2) = \frac{g(2)f'(2) - f(2)g'(2)}{[g(2)]^2} = \frac{(2)(-2) - (8)(1)}{2^2} = \frac{-12}{4} = \boxed{-3}$$

3. $h(x) = \frac{2f(x)}{3g(x)}, h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	7	4	4	5
2	-6	2	-1	-1
3	-3	-3	-2	-1
6	-2	5	-2	-4

$$h'(2) = \frac{2}{3} \left[\frac{g(2)f'(2) - f(2)g'(2)}{[g(2)]^2} \right] = \frac{2}{3} \left[\frac{(2)(-1) - (-6)(-1)}{2^2} \right] = \frac{2}{3} \left(\frac{-8}{4} \right) = \boxed{-\frac{4}{3}}$$

4. $h(x) = f(g(x)), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-7	6	-4	3
2	-8	2	-1	2
3	7	5	-4	5
6	7	3	2	-1

$$\begin{aligned} h'(x) &= f'(g(x))g'(x) & h'(2) &= f'(g(2))g'(2) \\ & & &= f'(2)[2] = (-1)(2) = \boxed{-2} \end{aligned}$$

5. $h(x) = f(3x), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-6	-2	-5	-3
2	8	3	3	-1
3	7	-5	5	-1
6	8	-6	-1	-3

$$h'(x) = 3f'(3x) \quad h'(2) = 3f'(6) = 3(-1) = \boxed{-3}$$

6. $h(x) = f(g(2x)), h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	6	2	-1	2
2	-6	3	-4	-2
3	1	-4	3	-1
6	4	-4	-5	2

$$h'(x) = f'(g(2x)) [g'(2x)](2) \quad h'(2) = f'(g(4)) (g'(2))(2) \\ = f'(3) [-2](2) = (3)(-4) = \boxed{-12}$$

7. $h(x) = g(f(2x)), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-7	2	-1	-5
2	3	2	-4	1
3	-6	-5	-5	4
4	3	4	-1	-2
6	3	-1	2	-4

$$h'(x) = g'(f(2x)) [f'(2x)](2) \quad h'(2) = g'(f(4)) (f'(4))(2) \\ = g'(3) [-1](2) = (4)(-2) = \boxed{-8}$$

8. $h(x) = f(x^2), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	6	2	5	-5
2	-5	3	4	-5
3	2	5	-1	5
4	-2	2	-2	-2
6	8	5	5	-4

$$h'(x) = f'(x^2) [2x] \quad h'(2) = f'(4) [4] \\ = (-2)(4) = \boxed{-8}$$

9. $h(x) = \frac{f(5x)}{g(x)}$, $h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-5	-6	-2	-1
2	-7	3	1	1
3	4	-3	-1	-5
6	2	-3	3	-4
10	-6	4	-2	2

$$h'(x) = \frac{g(x)[5f'(5x)] - [f(5x)](g'(x))}{[g(x)]^2} \quad h'(2) = \frac{g(2)[5f'(10)] - f(10)g'(2)}{[g(2)]^2}$$

$$= \frac{(3)(5)(-2) - (-6)(1)}{3^2} = \frac{-30 + 6}{9} = \boxed{-\frac{8}{3}}$$

10. $h(x) = f(3x)g(x)$, $h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	2	-1	1	1
2	8	3	-3	1
3	2	-1	4	5
6	2	-2	-2	4

$$h'(x) = f(3x)g'(x) + 3f'(3x)[g(x)] \quad h'(2) = f(6)g'(2) + 3f'(6)[g(2)]$$

$$= (2)(1) + 3(-2)(3) = \boxed{-16}$$

11. $h(x) = f(5x) - x^2$, $h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-1	3	3	-5
2	7	4	1	2
3	7	-4	2	-3
5	-4	-1	-2	-4
6	1	3	5	5

$$h'(x) = 5f'(5x) - 2x \quad h'(1) = 5f'(5) - 2(1) = 5(-2) - 2$$

$$= \boxed{-12}$$

12. $h(x) = g(f(2x)) - 2x$, $h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	5	2	4	-1
2	4	3	-3	-3
3	6	2	-4	-5
4	1	2	-3	-4
6	8	-4	-4	-5

$$h'(x) = g'(f(2x))[f'(2x)](2) - 2 \quad h'(2) = g'(f(4))[f'(4)](2) - 2$$

$$= g'(1)[(-3)](2) - 2 = (-1)(-3)(2) - 2 = \boxed{4}$$

13. $h(x) = \frac{f(2x)}{g(x)} - 5x, h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	7	-5	-4	-3
2	-5	3	5	-2
3	-6	-4	-4	-2
4	2	-4	4	1
6	-2	-1	-2	-3

$$h'(x) = \frac{g(x)[2f'(2x)] - f(2x)g'(x)}{[g(x)]^2} - 5 \quad h'(2) = \frac{g(2)[2f'(4)] - f(4)g'(2)}{[g(2)]^2} - 5$$

$$= \frac{(3)(2)(4) - (2)(-2)}{3^2} - 5 = \frac{28}{9} - 5 = \boxed{-\frac{17}{9}}$$

14. $h(x) = f(2x)g(4x), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	2	3	-3	-1
2	-6	3	-2	-3
3	-7	-3	4	3
4	-3	6	2	2
6	8	3	-5	-1
8	4	-3	4	3

$$h'(x) = f(2x)[4g'(4x)] + 2f'(2x)[g(4x)] \quad h'(2) = f(4)[4g'(8)] + 2f'(4)g(8)$$

$$= (-3)(4)(-3) + 2(2)(-3) = \boxed{-48}$$

15. $h(x) = \frac{f(2x)}{g(4x)}, h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	1	3	-1	5
2	-5	-6	-4	4
3	-3	2	1	5
4	7	2	4	-5
6	-8	-6	-4	-1

$$h'(x) = \frac{g(4x)[2f'(2x)] - f(2x)[4g'(4x)]}{[g(4x)]^2} = \frac{g(4)[2f'(2)] - f(2)[4g'(4)]}{[g(4)]^2}$$

$$= \frac{(2)(2)(-1) - (-5)(4)(-5)}{(2)^2} = \frac{-16 - 100}{4} = \boxed{-29}$$

16. $h(x) = f(g(x^2)), h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	2	-6	-1	3
2	4	3	3	-3
3	-4	1	-5	5
4	3	3	-4	-2
6	-1	-5	2	-5

$$h'(x) = f'(g(x^2))[g'(x^2)](2x) \quad h'(2) = f'(g(4))[g'(4)](4)$$

$$= f'(3)[-2](4) = (-5)(-8) = \boxed{40}$$

17. $h(x) = f(g(2x)), h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-1	2	-1	-2
2	-5	2	-3	1
3	4	-6	3	-1
6	-2	-4	4	-5

$$h'(x) = f'(g(2x)) [g'(2x)] (2) \quad h'(1) = f'(g(2)) [g'(2)] (2)$$

$$= f'(2) [1] (2) = (-3)(2) = \boxed{-6}$$

18. $h(x) = g(f(2x)) - 2x, h'(2) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	7	1	-2	3
2	1	3	5	-4
3	4	6	-2	-1
4	3	-5	-4	-3
6	4	5	3	-1

$$h'(x) = g'(f(2x)) [f'(2x)] (2) - 2 \quad h'(2) = g'(f(4)) [f'(4)] (2) - 2$$

$$= g'(3) [-4] (2) - 2 = (-1)(-8) - 2 = \boxed{6}$$

19. $h(x) = f(3x)g(4x), h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-3	2	-5	3
2	-3	-2	3	-2
3	3	-3	1	-5
4	1	-5	2	5
6	-7	-1	-3	3

$$h'(x) = f(3x) [4g'(4x)] + 3f'(3x) [g(4x)] \quad h'(1) = f(3) [4g'(4)] + 3f'(3) [g(4)]$$

$$= (3)(4)(5) + (3)(1)(-5) = \boxed{45}$$

20. $h(x) = \frac{f(4x)}{g(x)} - 3x^2, h'(1) =$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-4	-2	1	5
2	-1	1	3	-1
3	-2	1	4	5
4	6	6	1	3
6	6	6	-3	1

$$h'(x) = \frac{g(x) [4f'(4x)] - f(4x) [g'(x)]}{[g(x)]^2} - 6x \quad h'(1) = \frac{g(1) [4f'(4)] - f(4) g'(1)}{[g(1)]^2} - 6(1)$$

$$= \frac{(-2)(4)(1) - (6)(5)}{(-2)^2} - 6 = -\frac{38}{4} - 6 = \boxed{-\frac{31}{2}}$$